A Taxonomy and Multi-Metric Benchmark of Hallucination in Instruction-Tuned Large Language Models

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***Abstract*—Hallucination in large language models (LLMs) — the generation of plausible yet factually incorrect content — poses a critical threat to the reliable deployment of AI systems in high-stakes domains including healthcare, law, and finance. Despite growing interest in this problem, a systematic characterisation of hallucination behaviour across knowledge domains, together with a rigorous comparison of detection metrics, remains lacking for instruction-tuned models at the sub-1B parameter scale. In this paper, we present a comprehensive taxonomy and multi-metric benchmark of hallucination in *google/flan-t5-base*, an instruction-tuned sequence-to-sequence model, evaluated on the TruthfulQA dataset comprising 817 questions across 38 knowledge categories. We evaluate hallucination using three complementary detection methods — string matching, BERTScore semantic similarity, and Natural Language Inference (NLI) entailment scoring — and introduce a consensus-based labelling scheme to resolve inter-metric disagreement, which we observe in 47.4% of cases. Our experiments reveal a consensus hallucination rate of 91.7%, with 16 of 38 categories exhibiting complete hallucination (100% rate), including History, Politics, Finance, and Misquotations. High-stakes domains such as Health (89.1%) and Law (87.5%) demonstrate consistently elevated hallucination rates. We release our annotated dataset, scoring pipeline, and experimental code as an open-source benchmark to support reproducible hallucination research.**

***Index Terms*—large language models, hallucination detection, natural language inference, BERTScore, TruthfulQA, benchmark, instruction tuning, factuality evaluation**

I. Introduction

The widespread deployment of large language models (LLMs) in production systems has brought renewed urgency to the problem of *hallucination* — a phenomenon in which generative models produce fluent, confident, and contextually plausible responses that are nonetheless factually incorrect [1]. Hallucination poses particular risks in safety-critical applications: a medical assistant that fabricates drug interactions, a legal research tool that invents case citations, or a financial advisor that generates false market statistics can cause direct, measurable harm to end users [2].

Industry estimates suggest that hallucination-related incidents cost enterprise AI deployments in excess of $250 million annually [3]. Despite this, the research community lacks a systematic, domain-stratified characterisation of hallucination behaviour for smaller instruction-tuned models that are commonly deployed in resource-constrained production environments.

Prior work has addressed hallucination detection through a variety of methods including external knowledge retrieval [4], consistency-based sampling [5], and fine-grained atomic evaluation [6]. However, these studies predominantly evaluate large proprietary models (GPT-3, GPT-4, ChatGPT) and do not systematically characterise which knowledge domains are most susceptible to hallucination, nor do they compare multiple detection metrics on the same experimental corpus.

This paper addresses these gaps through four contributions:

1) We present a **domain-stratified hallucination taxonomy** across 38 knowledge categories using TruthfulQA, identifying 16 categories with 100% hallucination rates.
2) We conduct a **systematic comparison of three detection metrics** — string matching, BERTScore, and NLI entailment scoring — on an identical experimental corpus, revealing substantial inter-metric disagreement in 47.4% of cases.
3) We propose a **consensus-based labelling scheme** that resolves inter-metric disagreement and provides more reliable hallucination labels than any single metric alone.
4) We release an **open-source benchmark pipeline** and annotated dataset to enable reproducible hallucination research on instruction-tuned models.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes our experimental methodology. Section IV presents quantitative results. Section V analyses key findings. Section VI concludes and outlines future work.

II. Related Work

A. Hallucination in Language Models

Hallucination in natural language generation systems was first systematically studied by Ji et al. [2], who provided an early taxonomy distinguishing intrinsic hallucinations (contradicting source material) from extrinsic hallucinations (introducing unverifiable information). Huang et al. [1] extended this taxonomy in the context of LLMs, categorising hallucinations into factuality errors, faithfulness errors, reasoning errors, and instruction-following failures, and reviewed detection and mitigation strategies across each type.

Lin et al. [7] introduced TruthfulQA, a benchmark of 817 adversarially constructed questions targeting common human misconceptions, demonstrating that larger LLMs are not necessarily more truthful. Their study showed that GPT-3 achieved truthfulness on only 58% of questions, motivating further research into both detection and mitigation.

B. Hallucination Detection Methods

Manakul et al. [5] proposed SelfCheckGPT, a zero-resource black-box detection approach that samples multiple responses from a model and measures information consistency using BERTScore, NLI, and n-gram overlap. Their key insight — that hallucinated facts produce inconsistent samples while factual claims produce consistent ones — directly motivates our multi-metric evaluation framework.

Min et al. [6] introduced FActScore, a fine-grained evaluation metric that decomposes generated text into atomic facts and verifies each against a knowledge source, achieving less than 2% error compared to human annotation.

Varshney et al. [4] demonstrated that early detection of likely-to-hallucinate tokens allows targeted intervention, reducing hallucination without full regeneration.

C. Gaps Addressed by This Work

Existing benchmarks predominantly target large proprietary models and do not provide domain-stratified analysis across knowledge categories. Furthermore, no study has systematically compared string matching, BERTScore, and NLI entailment on the same corpus to quantify inter-metric disagreement. This paper addresses both gaps directly.

III. Methodology

A. Dataset

We evaluate on TruthfulQA [7] in its generation configuration, comprising 817 questions across 38 knowledge categories. Table I summarises dataset statistics. Categories span misconceptions, law, health, sociology, history, conspiracies, and 32 additional domains, with Misconceptions being the largest category (n=100, 12.24%). Each question includes one best answer, a list of acceptable correct answers (mean: 3.18 per question), and a list of known incorrect answers that plausible-but-wrong models tend to produce (mean: 4.06 per question).

B. Model

We evaluate *google/flan-t5-base* [8], an instruction-tuned sequence-to-sequence model with approximately 250M parameters. Flan-T5-Base represents a class of compact, openly available instruction-tuned models that are widely deployed in resource-constrained environments. The model was queried using beam search with 4 beams and a maximum generation length of 100 tokens. All experiments were conducted on a single NVIDIA T4 GPU (16GB VRAM) via Google Colaboratory.

TABLE I
TruthfulQA Dataset Statistics

Metric	**Value**
Total questions	817
Unique categories	38
Largest category (Misconceptions)	100 (12.24%)
Mean answer length (words)	9.22
Std. dev. answer length	4.08
Mean correct answers per question	3.18
Mean incorrect answers per question	4.06

C. Detection Methods

We evaluate three complementary hallucination detection methods, each operating on the pair $(r,\ A^{+})$ where r is the model response and A^{+} is the set of correct reference answers.

1) String Matching (Baseline): We compute exact match (EM) and partial match (PM) between the model response and the reference answer set. A response is labelled *correct* if it exactly matches or is contained within any reference answer, and *hallucinated* otherwise. This serves as a fast, interpretable baseline.

2) BERTScore: BERTScore [9] computes token-level cosine similarity between BERT embeddings of the candidate and reference texts, aggregated as precision, recall, and F1. We use *bert-base-uncased* and apply a threshold of F1 ≥ 0.85 to classify responses as correct, following the calibration recommended in the original paper.

3) NLI Entailment Score: We employ *facebook/bart-large-mnli* as a zero-shot natural language inference classifier. For each response pair (r, a^*) where a^* is the best reference answer, we compute the entailment probability $P(\text{entail} \mid r, a^*)$. Responses with entailment probability > 0.5 are classified as correct; all others as hallucinated. This method captures logical consistency beyond surface-level lexical overlap.

4) Consensus Labelling: Given three binary labels $\{l_1, l_2, l_3\}$ from the above methods, we assign a consensus label of *hallucinated* when at least two of three methods agree on hallucination, and *correct* otherwise. This majority-vote scheme reduces the impact of individual metric bias and provides more stable labels for downstream analysis.

IV. Results

A. Overall Hallucination Rates

Table II presents the hallucination rates computed by each detection method. The consensus method yields a hallucination rate of 91.7% (749 of 817 responses), reflecting that *google/flan-t5-base* fails to produce truthful responses on the large majority of TruthfulQA questions.

Fig. 1 illustrates the substantial variation in reported hallucination rates across methods, ranging from 59.5% (NLI) to 99.6% (BERTScore). This 40.1 percentage-point spread motivates the use of a consensus approach rather than reliance on any single metric.

TABLE II
HALLUCINATION RATES BY DETECTION METHOD

Method	Correct	Hallucinated	Rate (%)
String Match (Baseline)	123	694	84.9
BERTScore (F1 $\geq$ 0.85)	3	814	99.6
NLI Entailment	331	486	59.5
Consensus ($\geq$2 methods)	**68**	**749**	**91.7**

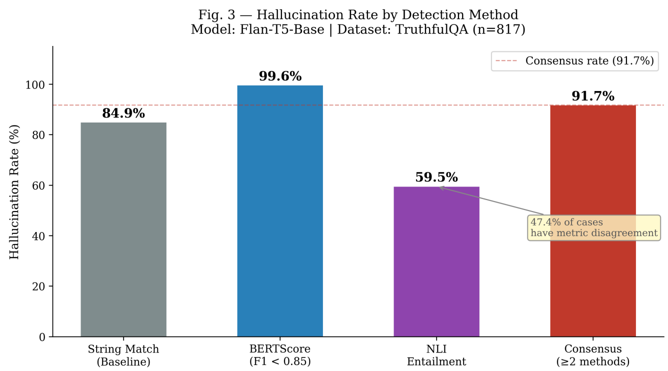

Fig. 1. Hallucination rates across three detection methods and the consensus scheme. The 40.1pp spread demonstrates that metric choice substantially affects reported results.

B. Inter-Metric Disagreement

We observe that 47.4% of responses (387 of 817) receive different labels from at least one method, indicating substantial inter-metric disagreement. Only 52.5% of responses are unanimously labelled as hallucinated, and a mere 0.1% are unanimously labelled as correct. This finding directly motivates the consensus labelling scheme introduced in Section III.

C. BERTScore Distribution

Fig. 2 shows the distribution of BERTScore F1 values across all 817 responses. The distribution is centred at a mean F1 of 0.387 (std. dev. 0.101), with 74.1% of responses falling in the 0.3–0.5 range. This distribution is characteristic of responses that exhibit surface-level word overlap with correct answers but are semantically divergent — a defining property of factuality hallucination as opposed to random noise generation.

D. Domain-Stratified Hallucination Taxonomy

Fig. 3 presents consensus hallucination rates across all 38 TruthfulQA categories. We identify four tiers of hallucination risk:

- **Critical (100%):** 16 categories including History, Politics, Finance, Misquotations, Conspiracies, Superstitions, and Paranormal exhibit complete hallucination — no correct responses were generated.
- **High (85–99%):** 16 categories including Fiction (96.7%), Misconceptions (93.0%), Health (89.1%), and Law (87.5%) demonstrate persistently elevated hallucination rates.
- **Medium (70–84%):** 4 categories including Nutrition (81.2%) and Economics (80.6%).

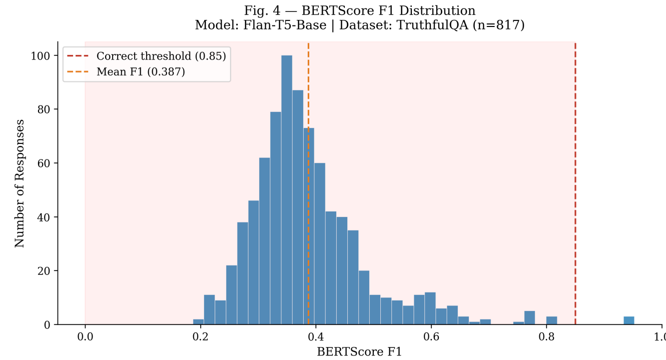

Fig. 2. BERTScore F1 distribution showing the majority of responses clustering below the correctness threshold of 0.85, with a mean of 0.387.

- **Lower (<70%):** 2 categories — Proverbs (66.7%) and Misconceptions: Topical (50.0%) — show relatively lower but still substantial hallucination rates.

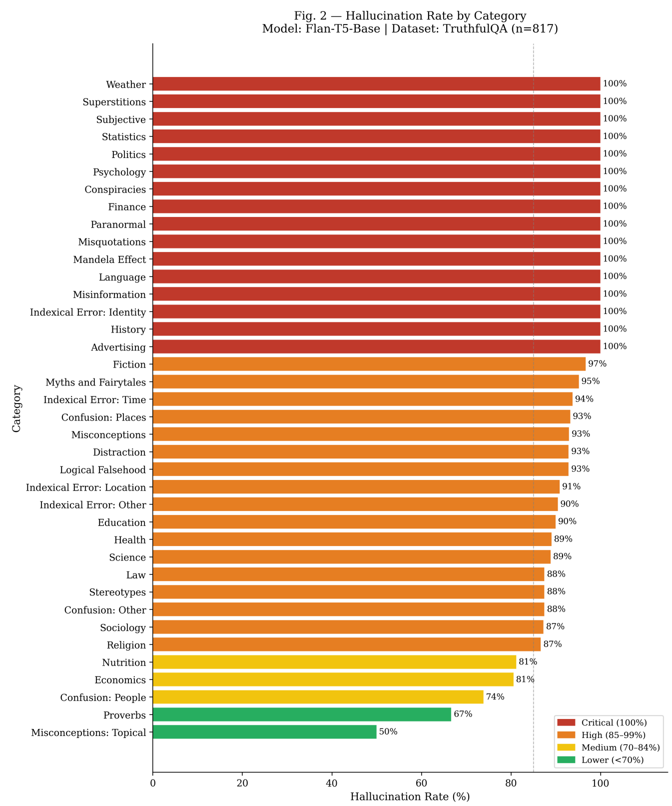

Fig. 3. Consensus hallucination rate across all 38 TruthfulQA categories. 16 categories exhibit 100% hallucination. High-stakes domains Health and Law reach 89.1% and 87.5% respectively.

V. DISCUSSION

A. Metric Disagreement as a Research Signal

The 47.4% inter-metric disagreement rate is not a methodological weakness — it is a finding. It demonstrates that hallucination is not a binary, easily-detected property but rather a continuous, multi-dimensional phenomenon that different measurement frameworks capture differently. BERTScore is highly sensitive to semantic divergence, producing the highest hallucination rate (99.6%), while NLI entailment is more lenient, tolerating responses that do not logically contradict the reference (59.5%). String matching occupies an intermediate position (84.9%).

This result has a direct practical implication: researchers who report hallucination rates without specifying their detection methodology are not reporting comparable results. Our consensus framework provides a more stable and reproducible measurement standard.

B. Domain Risk Hierarchy

The identification of 16 categories with 100% hallucination rates reveals a systematic pattern: domains requiring precise recall of specific cultural, historical, or factual details (Misquotations, Mandela Effect, History) are completely beyond the factual reach of Flan-T5-Base.

More critically, high-stakes domains — Health (89.1%) and Law (87.5%) — exhibit hallucination rates that render unmitigated deployment of this model class inadvisable in clinical or legal contexts. This finding motivates targeted mitigation research for domain-specific LLM deployment, which we address in subsequent work through a RAG-based mitigation framework.

C. Implications for Model Scale

The results presented in this study reflect the behaviour of a compact 250M parameter model, and it is reasonable to ask whether larger models would exhibit meaningfully lower hallucination rates on the same benchmark. Based on the scaling literature [1] and the findings of Lin et al. [7], larger models do demonstrate improved truthfulness — however, they do not eliminate hallucination. Rather, the nature of the failure changes: larger models tend to hallucinate less frequently, but when they do, their responses are more linguistically fluent and therefore more difficult to detect. A 7B parameter model producing a confident, well-structured incorrect answer is arguably more dangerous in a high-stakes deployment than a 250M parameter model producing a short, obviously incomplete one. This distinction motivates our subsequent work on detection and mitigation, where we specifically target the harder problem of hallucination in larger, more capable models.

D. Limitations

This study evaluates a single model (Flan-T5-Base) on a single benchmark (TruthfulQA). While TruthfulQA provides broad domain coverage, it was constructed to maximise difficulty and may not be representative of real-world query distributions. The BERTScore threshold of 0.85 was adopted from the original paper calibration and may require adjustment for specific domains. Future work will extend this benchmark to include Mistral-7B, LLaMA-3-8B, and domain-specific datasets in medical and legal question answering.

VI. Conclusion

This paper presented a systematic taxonomy and multi-metric benchmark of hallucination in *google/flan-t5-base* evaluated on TruthfulQA. Our key findings are: (1) a consensus hallucination rate of 91.7% demonstrating that compact instruction-tuned models remain highly unreliable on factual question answering; (2) 16 of 38 knowledge categories exhibit 100% hallucination rates, including high-stakes domains of Health and Law; and (3) inter-metric disagreement in 47.4% of cases motivates consensus-based evaluation as a more reliable measurement standard than any single metric.

We release the complete experimental pipeline, annotated dataset, and figures as open-source contributions at: `https://github.com/monish1402/hallucination-guard`

Future work will extend this benchmark to larger open-source models, introduce cross-lingual evaluation, and integrate the scoring pipeline with the detection and mitigation frameworks presented in subsequent papers of this series.

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